Week 6-3

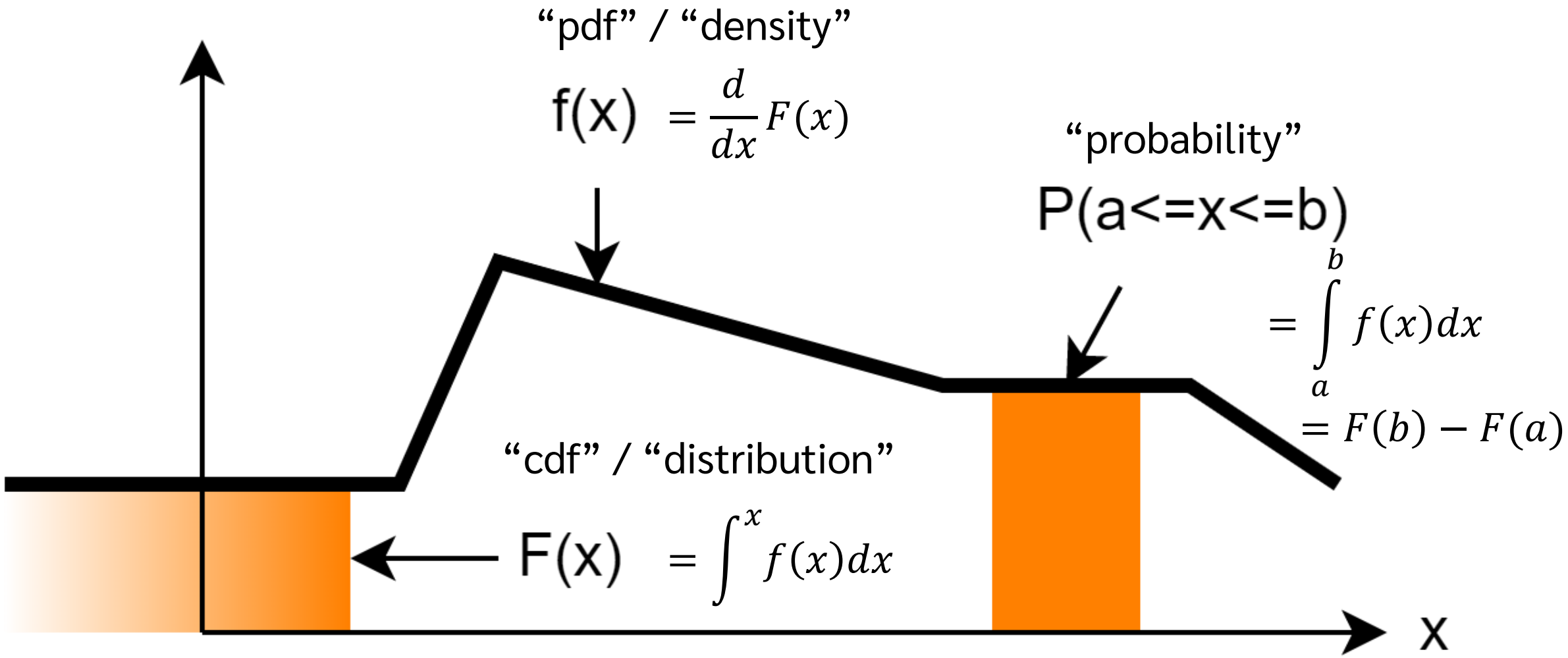
Continuous Random Variable 2

September 12, 2024

2110205 STATS COMP ENG

CHULA
COMPUTER
ΣNGINEERING

RECAP: Continuous random variable



Mean of Continuous Random Variable

ATTENDANCE CHECK

Question. What is the $E(X)$ if X is a continuous random variable with distribution $U(2,4)$.

1. 2
2. 3
3. 4

Transformation of Random Variables – Ex. 1

Problem. Given random variable X with distribution $U(0,2)$. Find range, cdf, and pdf of $Y = X^2$. ($F_X(x) = \frac{x}{2}$, $f_X(x) = \frac{1}{2}$ in range $[0,2]$)

Ans. Range of Y is [0,4]. To get cdf, we can do:

$$F_Y(y) = P(Y \leq y) = P(X^2 \leq y) = P(X \leq \sqrt{y}) = F_X(\sqrt{y}) = \boxed{\frac{\sqrt{y}}{2}}.$$

So we get $F_Y(y) = \frac{\sqrt{y}}{2}$. To get pdf $f_Y(y) = \frac{d}{dy} F_Y(y) = \frac{d}{dy} \frac{\sqrt{y}}{2} = \boxed{\frac{1}{4\sqrt{y}}}.$

Cont'd.

Alternative (easier?) answer:

We can also do a **change of variable.**

Start with: $y = x^2$ ➔ $dy = 2xdx$ ➔ $\frac{dy}{2x} = dx$ ➔ $\frac{dy}{2\sqrt{y}} = dx.$

Then: $f_X(x)dx = \frac{1}{2}dx = \frac{dy}{4\sqrt{y}} = f_Y(y)dy.$

Therefore: $f_Y(y) = \frac{1}{4\sqrt{y}}$

Example 2

Problem. Let X be a distribution $\exp(\lambda)$. Find PDF of $Y = X^2$.

Ans. Use change of variable.

Start with: $y = x^2$ ➔ $dy = 2xdx$ ➔ $\frac{dy}{2x} = dx$ ➔ $\frac{dy}{2\sqrt{y}} = dx$.

Then:

$$f_X(x)dx = \lambda e^{-\lambda x}dx = \frac{\lambda e^{-\lambda\sqrt{y}}}{2\sqrt{y}}dy$$

Thus: $f_Y(y) = \frac{\lambda e^{-\lambda\sqrt{y}}}{2\sqrt{y}}$.

Expectation

As with discrete random variable, we can calculate **Expection** or **mean** of the random variable:

$$\mu = E(X) = \int_a^b x f(x) dx.$$

This is extremely similar to Discrete's:

$$E(X) = \sum x_i p(x_i)\,.$$

Properties of Expectation

Same as discrete:

1. $E(X+Y) = E(X) + E(Y)$
2. $E(aX+b) = aE(X) + b$ when a and b is constant.

Example 3

Problem. Let $X \sim U(0,2)$. Find $E(X)$.

Ans. $f_X(x) = 0.5$.

So $E(X) = \int_0^2 f_X(x)xdx = \frac{1}{2}\int_0^2 xdx = \frac{x^2}{4}\Big|_0^2 = 1.$

Expectation of Function

Same as discrete again:

$$E\big(h(X)\big) = \int_{-\infty}^{\infty} h(x)f(x)dx.$$

Example 4

Problem. Given $X \sim \exp(\lambda)$. Find $E(X^2)$.

Ans.

$$E(X^2) = \int_0^\infty x^2 \lambda e^{-\lambda x} dx = \cdots = \frac{2}{\lambda^2}$$

(We will look at the integration in detail next class)

Quiz 1

Problem. Suppose $X \sim U(0,1)$. Find pdf of $Y = X^2$.

10 MIN BREAK

Variance of Continuous Random Variable

Variance

All formula and properties is the same as discrete's:

Formula: $Var(X) = E\left((X-\mu)^2\right)$.

Properties:

1. If X and Y is independent: $Var(X+Y) = Var(X) + Var(Y)$.
2. For constants a and b: $Var(aX+b) = a^2 Var(x)$.
3. $Var(X) = E(X^2) - E(X)^2$.

Example 5

Problem. Let $X \sim U(0,1)$. Find $Var(X)$.

Ans. We know $\mu = 0.5$.

So: $E\left((X-\mu)^2\right) = \int_0^1 \left(x - \frac{1}{2}\right)^2 dx$. Give $u = x - \frac{1}{2}, \frac{du}{dx} = 1$.

We get $\int_{-\frac{1}{2}}^{\frac{1}{2}} u^2 du = \frac{u^3}{3}\Big|_{-\frac{1}{2}}^{\frac{1}{2}} = \frac{\left(\frac{1}{2}\right)^3}{3} - \frac{\left(-\frac{1}{2}\right)^3}{3} = \frac{1}{24} + \frac{1}{24} = \frac{1}{12}$

Example 6

Problem. Find the mean and variance of $N(0,1)$.

Ans. By definition, $\mu = 0$ and $\sigma^2 = 1$.

(We will look at the integration in detail next class)

Mean and Variance of Distributions

	pdf	cdf	Range	Mean	Variance
Normal	$\frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/2\sigma}$	Use table	$[-\infty, \infty]$	μ	σ^2
Exponential	$\lambda e^{-\lambda x}$	$1 - e^{-\lambda x}$	$[0, \infty]$	$1/\lambda$	$1/\lambda^2$
Uniform	$\frac{1}{b-a}$	$\frac{x-a}{b-a}$	$[a, b]$	$\frac{a+b}{2}$	$\frac{1}{12}(a-b)$

Quantiles of Continuous Random Variable

Median of Random Variable

Median of the random variable:

$$x \text{ when } P(X \leq x) = 0.5 \text{ --OR-- } x \text{ when } F(x) = 0.5.$$

Example: Find median of $X \sim \exp(\lambda)$.

Ans. $F(x) = 1 - e^{-\lambda x} = 0.5;\ \ x = \ln(2)/\lambda$.

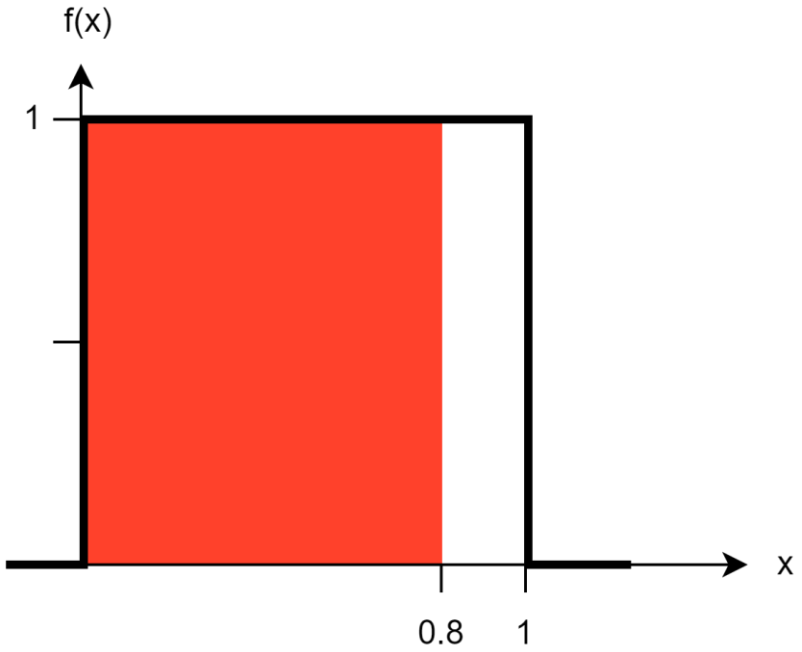
Quantile

pth-quantile of random variable X is q_p where $F(q_p) = p$.

(e.g. median is $q_{0.5}$)

Example: Find $q_{0.8}$ of $U(0,1)$.

Ans. 0.8



Percentiles/Deciles/Quartiles

- **Percentile:** 100 steps. E.g. 75th percentile is $q_{0.75}$.
- **Deciles:** 10 steps. E.g. 8th decile is $q_{0.8}$.
- **Quartiles:** 4 steps. E.g. 1st quartile is $q_{0.25}$.

Quiz 2

1) What is range and mean of $X + X$ when $X \sim U(0,2)$.

Range = [0,4] E(X)=1 E(X+X)=2

2) Let $X \sim N(50,100)$. Calculate $P(40 \leq x \leq 60)$.

Let $Y = \frac{X-50}{10}$. Then, $Y \sim N(0,1)$. Thus, we get $P(-1 \leq Y \leq 1) = 0.68$.

3) What is $q_{0.3}$ of $U(5,15)$.

$= 0.3 \times (15 - 5) + 5 = 8$

END OF TODAY LECTURE

NEXT TIME >>

- Review of continuous random variable.
- Proof of all distribution formula.
- More complex calculus.
- More applied problems.

Homework

- **Homework 6 is OUT NOW.**
- **Deadline:** Wednesday, September 18.
  - Late submission up to 24 hours (with reduced points)
  - Solution will be posted on Friday.